

A Simplified Replication of Rust’s Dynamic Bus Engine Replacement Model

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May 1, 2026

Abstract

This paper presents a simplified replication of the dynamic discrete-choice model in [Rust \(1987\)](#). The original paper studies Harold Zurcher’s monthly decision to keep or replace bus engines in the Madison Metropolitan Bus Company fleet. The replacement decision is dynamic because replacing an engine is costly today, while keeping an older engine increases current and future maintenance costs. This replication estimates a regenerative optimal replacement model with a linear maintenance-cost function, discrete mileage transitions, Type-I extreme value choice shocks, and a fixed discount factor. The model is estimated using a nested fixed point algorithm in which the inner loop solves the dynamic programming problem and the outer loop maximizes the likelihood. The empirical analysis considers three samples: bus groups 1–3, group 4, and groups 1–4 pooled. The estimated model is used to construct replacement hazards, ex-ante value functions, equilibrium mileage distributions, and counterfactual expected replacement demand functions with respect to the replacement-cost parameter. The results reproduce the main economic logic of Rust’s model: replacement probabilities increase with mileage, higher replacement costs reduce replacement demand, and pooling heterogeneous bus groups can distort the estimated hazard and counterfactual predictions.

1 Introduction

Rust’s bus engine replacement paper is one of the canonical applications of dynamic discrete-choice estimation. The economic object is a monthly decision: whether Harold Zurcher, the superintendent of maintenance at Madison Metro, keeps the current bus engine or replaces

it. Replacement is costly, but keeping an old engine raises current and future operating costs. The model therefore formalizes a dynamic trade-off between the immediate cost of replacement and the future cost consequences of continued use.

The original paper is important for two reasons. First, it provides a “bottom-up” structural model of replacement investment. Instead of starting from an aggregate capital-stock depreciation equation, the model derives replacement investment from individual decisions over specific capital goods. Second, it introduces the nested fixed point algorithm for estimating dynamic discrete-choice models. In the inner loop, the researcher solves the dynamic programming problem for a candidate structural parameter vector. In the outer loop, the researcher searches over the structural parameters to maximize the likelihood.

This paper implements a simplified replication of Rust’s model in Python. The replication keeps the main dynamic structure of the original paper: a binary keep-replace choice, a mileage state, stochastic mileage transitions, replacement as regeneration, unobserved choice-specific shocks, and a likelihood that is computed from the solution to the dynamic programming problem. To keep the replication transparent, the empirical specification focuses on a linear maintenance-cost function,

$$c(x) = \theta x,$$

and fixes the discount factor at

$$\beta = 0.9999.$$

The replication has five objectives. First, it reconstructs the dynamic replacement model in a clear and computable way. Second, it estimates the model separately for bus groups 1–3, group 4, and the pooled sample of groups 1–4. Third, it evaluates the fitted model using replacement hazards and value functions. Fourth, it computes equilibrium mileage distributions implied by the estimated policy. Fifth, it conducts counterfactual replacement-cost experiments and constructs expected replacement demand functions.

2 Rust’s Original Framework

2.1 Economic Environment

Rust models bus engine replacement as a regenerative optimal stopping problem. The state variable is accumulated mileage since the last replacement. Each period, the decision maker chooses whether to continue operating the current engine or to replace it. If the engine is kept, the bus remains in the current state and mileage evolves stochastically. If the engine is replaced, the system regenerates because the state is reset to the new-engine mileage

distribution.

The key economic trade-off is simple. Keeping the engine avoids the replacement cost today but exposes the firm to higher expected maintenance and breakdown costs in the future. Replacing the engine requires a large current cost but lowers future expected operating costs. Thus, observed replacement behavior contains information about the decision maker's perceived maintenance-cost function and replacement-cost trade-off.

2.2 Why a Dynamic Model Is Needed

A static model would compare only current replacement cost and current maintenance cost. However, replacement decisions are forward-looking. A bus engine that is kept today will continue to be used in the future, and its future state distribution depends on today's decision. This is why the dynamic replacement probability depends not only on current maintenance cost but also on the continuation value associated with each action.

The dynamic model also avoids a degenerate deterministic threshold rule by introducing unobserved choice-specific state variables. These shocks are observed by the decision maker but not by the econometrician. As a result, replacement is probabilistic from the econometrician's perspective, even though the decision maker follows an optimal policy given the full state.

2.3 Nested Fixed Point Estimation

The nested fixed point algorithm is the central computational contribution. For each parameter vector, the value function must be solved before the likelihood can be evaluated. Estimation therefore nests the dynamic programming solution inside the likelihood maximization problem. In the original paper, Rust uses this algorithm to estimate structural parameters of the cost function and transition process.

The present replication follows this same logic. The inner loop solves the Bellman fixed point. The outer loop chooses the replacement-cost parameter, maintenance-cost parameter, and mileage transition probabilities to maximize the sample likelihood.

3 Data and State Space

3.1 Original Data

The original data consist of monthly maintenance records for buses in the Madison Metro fleet. The relevant variables for this simplified replication are the bus identifier, month,

accumulated mileage since the last engine replacement, and the replacement decision. The original paper considers multiple bus groups, but the main empirical analysis focuses on groups 1–4 because these buses have relatively homogeneous utilization patterns and available replacement-cost information.

This replication follows the paper-style grouping and estimates three samples:

1. groups 1, 2, and 3;
2. group 4;
3. groups 1–4 pooled.

The separate samples allow the structural parameters to differ between newer buses and the older group 4 buses. The pooled model imposes common parameters across all groups.

3.2 Mileage State

Mileage is discretized into 90 states with 5,000-mile bins. Let

$$x_t \in \mathcal{X} = \{0, 1, \dots, 89\}$$

denote the mileage state in month t . The mileage level corresponding to state x is approximately

$$5000x$$

miles since the last replacement. State 0 corresponds to a new or recently replaced engine. The last state, state 89, is top-coded and should be interpreted as the highest mileage bin rather than as an exact mileage value.

3.3 Mileage Increments

Monthly mileage increments are discretized into three categories:

$$\Delta_t \in \{0, 1, 2\}.$$

The transition probabilities are

$$\mathbb{P}(\Delta_t = 0) = p_0, \quad \mathbb{P}(\Delta_t = 1) = p_1, \quad \mathbb{P}(\Delta_t = 2) = p_2,$$

where

$$p_0 + p_1 + p_2 = 1.$$

The category $\Delta_t = 2$ should be interpreted as moving two or more 5,000-mile bins in a month. In the Python implementation, usage increments are clipped at 2, so all large monthly mileage changes are grouped into this top category.

3.4 Replacement Decision

Let

$$i_t \in \{0, 1\}$$

denote the monthly replacement decision:

$$i_t = \begin{cases} 0, & \text{keep the current engine,} \\ 1, & \text{replace the current engine.} \end{cases}$$

The model explains the joint behavior of mileage states and replacement choices.

4 Model

4.1 Choice-Specific Utility

The decision maker observes the mileage state x_t and chooses whether to keep or replace the engine. Utility is written as the negative of cost plus an unobserved choice-specific shock. The deterministic maintenance-cost function is linear:

$$c(x_t; \theta) = \theta x_t,$$

where $\theta > 0$ measures the slope of maintenance cost with respect to the mileage state.

If the engine is kept, current deterministic utility is

$$u_0(x_t; \theta) = -c(x_t; \theta) = -\theta x_t.$$

If the engine is replaced, the decision maker pays replacement cost RC and starts next period with a new engine. Because $c(0; \theta) = 0$, the deterministic replacement utility is

$$u_1(x_t; RC) = -RC.$$

The realized utilities are

$$U_j(x_t) = u_j(x_t) + \varepsilon_t(j), \quad j \in \{0, 1\},$$

where $\varepsilon_t(0)$ and $\varepsilon_t(1)$ are choice-specific unobserved shocks.

4.2 Transition Law

If the engine is kept, next period's mileage state is

$$x_{t+1} = \min\{x_t + \Delta_t, 89\}.$$

If the engine is replaced, the engine regenerates and next period's mileage state is

$$x_{t+1} = \min\{\Delta_t, 89\}.$$

The replacement action therefore resets the state to the new-engine transition distribution.

Let $P_0(y \mid x)$ denote the transition probability from state x to state y when the engine is kept, and let $P_1(y \mid x)$ denote the transition probability when the engine is replaced. Then

$$P_0(y \mid x) = \sum_{k=0}^2 p_k \mathbf{1}\{y = \min(x + k, 89)\},$$

and

$$P_1(y \mid x) = \sum_{k=0}^2 p_k \mathbf{1}\{y = \min(k, 89)\}.$$

The transition matrix under replacement is independent of the current state because replacement regenerates the engine.

4.3 Conditional Independence

The empirical model relies on the standard conditional independence structure in Rust's dynamic discrete-choice framework. Conditional on the observed state and current action, the next observed state evolves independently of current unobserved utility shocks. This assumption permits the econometrician to solve the dynamic programming problem on the observed state space and then integrate over the extreme-value shocks to obtain closed-form dynamic logit choice probabilities.

4.4 Dynamic Programming Problem

Let $W(x)$ denote the ex-ante value function before the realization of the choice-specific shocks. Define the choice-specific value functions:

$$v_0(x) = -\theta x + \beta \sum_{y \in \mathcal{X}} P_0(y | x) W(y),$$

and

$$v_1(x) = -RC + \beta \sum_{y \in \mathcal{X}} P_1(y | x) W(y).$$

Since the replacement transition is regenerative,

$$\sum_{y \in \mathcal{X}} P_1(y | x) W(y) = \sum_{y \in \mathcal{X}} P_1(y | 0) W(y).$$

Therefore, the continuation value from replacement does not depend on the pre-replacement mileage state except through the current replacement cost.

Under Type-I extreme value shocks, the ex-ante value function satisfies the log-sum Bellman equation:

$$W(x) = \log [\exp\{v_0(x)\} + \exp\{v_1(x)\}].$$

The conditional replacement probability is

$$\mathbb{P}(i_t = 1 | x_t = x) = \frac{\exp\{v_1(x)\}}{\exp\{v_0(x)\} + \exp\{v_1(x)\}}.$$

The conditional keep probability is

$$\mathbb{P}(i_t = 0 | x_t = x) = \frac{\exp\{v_0(x)\}}{\exp\{v_0(x)\} + \exp\{v_1(x)\}}.$$

4.5 Economic Interpretation

The replacement probability increases with mileage when the maintenance-cost slope is positive. At low mileage, keeping the engine is attractive because maintenance costs are small and replacement is expensive. At high mileage, the current and future costs of continued use rise, so replacement becomes more likely. The exact shape of the hazard depends on the replacement cost, the maintenance-cost slope, the transition probabilities, and the discount factor.

5 Estimation

5.1 Parameter Vector

The structural parameter vector is

$$\Theta = (RC, \theta, p_0, p_1, p_2).$$

The discount factor is fixed at

$$\beta = 0.9999.$$

The estimation uses transformed parameters to impose the required constraints. The replacement cost and maintenance slope are exponentiated:

$$RC = \exp(\phi_{RC}), \quad \theta = \exp(\phi_{\theta}).$$

The transition probabilities are constructed using a softmax transformation:

$$p_k = \frac{\exp(z_k)}{\sum_{j=0}^2 \exp(z_j)},$$

with one of the three z_j normalized to zero.

5.2 Likelihood Function

The likelihood has two parts: the choice likelihood and the mileage-transition likelihood. Let

$$P_1(x; \Theta) = \mathbb{P}(i_t = 1 \mid x_t = x; \Theta)$$

and

$$P_0(x; \Theta) = \mathbb{P}(i_t = 0 \mid x_t = x; \Theta).$$

The choice log likelihood is

$$\ell_{\text{choice}}(\Theta) = \sum_t [i_t \log P_1(x_t; \Theta) + (1 - i_t) \log P_0(x_t; \Theta)].$$

Let n_k denote the number of observed monthly usage increments in category k . The transition log likelihood is

$$\ell_{\text{transition}}(\Theta) = \sum_{k=0}^2 n_k \log p_k.$$

The total log likelihood is

$$\ell(\Theta) = \ell_{\text{choice}}(\Theta) + \ell_{\text{transition}}(\Theta).$$

5.3 Nested Fixed Point Algorithm

The estimation uses a nested fixed point algorithm. For each candidate parameter vector, the inner loop solves the Bellman equation for $W(x)$. The outer loop chooses the parameter vector that maximizes the likelihood.

The implementation proceeds as follows:

1. Choose a trial parameter vector Θ .
2. Construct the maintenance-cost vector $c(x) = \theta x$.
3. Construct the keep transition matrix P_0 and replacement transition matrix P_1 from (p_0, p_1, p_2) .
4. Solve the log-sum Bellman equation for $W(x)$.
5. Compute the dynamic logit probabilities $\mathbb{P}(i_t = 1 \mid x_t)$ and $\mathbb{P}(i_t = 0 \mid x_t)$.
6. Evaluate the choice likelihood and transition likelihood.
7. Update the trial parameter vector using the outer-loop optimizer.

The Python implementation uses value iteration as the primary inner-loop solver. If value iteration does not converge, the code switches to a nonlinear root solver. The outer-loop optimization uses L-BFGS-B with bounded transformed parameters. The code also uses the previous value-function solution as a warm start to speed up the next fixed-point calculation.

5.4 Equilibrium Distribution

After estimation, the replacement policy implies a controlled transition matrix:

$$P^*(y \mid x) = \mathbb{P}(i = 0 \mid x)P_0(y \mid x) + \mathbb{P}(i = 1 \mid x)P_1(y \mid x).$$

The stationary mileage distribution π solves

$$\pi(y) = \sum_x \pi(x)P^*(y \mid x).$$

The expected monthly replacement rate is

$$D = \sum_x \pi(x) \mathbb{P}(i = 1 \mid x).$$

This object is an equilibrium replacement rate because both the state distribution and the policy are determined by the estimated dynamic model.

6 Estimation Results

6.1 Structural Parameter Estimates

Table 1 reports the main structural estimates.

Table 1: Dynamic NFXP estimation results for the linear-cost specification

Sample	RC	θ	p_0	p_1	p_2	Log likelihood	Obs.	Buses
Groups 1,2,3	26.2689	0.0127	0.3149	0.6793	0.0058	-2724.646	3931	67
Group 4	66.0078	0.0247	0.4059	0.5847	0.0094	-3422.050	4329	37
Groups 1–4 pooled	13.1709	0.0035	0.3286	0.6675	0.0039	-6112.062	8260	104

Notes: The table reports estimates from the simplified dynamic replacement model with linear maintenance cost $c(x) = \theta x$ and fixed discount factor $\beta = 0.9999$. The reported replacement-cost and maintenance-cost parameters are in the utility scale of the logit shocks, not directly in dollars.

The estimates imply substantial heterogeneity across samples. The estimated replacement-cost parameter is much larger for group 4 than for groups 1–3. The pooled model produces a much smaller replacement-cost estimate and a much flatter maintenance-cost slope. This is important because the pooled model averages different replacement patterns across bus groups.

The transition estimates show that most monthly mileage changes correspond to one 5,000-mile bin or no bin. For groups 1–3, the estimated probability of moving one bin is 0.6793, while the probability of moving two or more bins is only 0.0058. For group 4, the probability of no bin movement is larger and the probability of one-bin movement is smaller. These differences in usage dynamics affect the stationary distribution and the counterfactual demand function.

6.2 Dollar Scaling of Maintenance Slopes

The parameters in Table 1 are in utility units. A useful interpretation is obtained by scaling them with observed replacement-cost data from the original paper. The original replacement-

cost table reports average replacement costs of approximately \$9,499 for groups 1–3 and \$7,513 for group 4. Using these observed dollar costs as normalizations, one can compute an approximate dollar maintenance-cost slope per 5,000-mile state.

Table 2: Approximate dollar-scaled maintenance-cost slopes

Sample	Observed replacement cost	Estimated RC	Dollar scale	Implied dollar slope per 5,000 miles
Groups 1,2,3	\$9,499	26.2689	361.61	\$4.59
Group 4	\$7,513	66.0078	113.82	\$2.81
Groups 1–4 pooled	\$8,062	13.1709	612.12	\$2.14

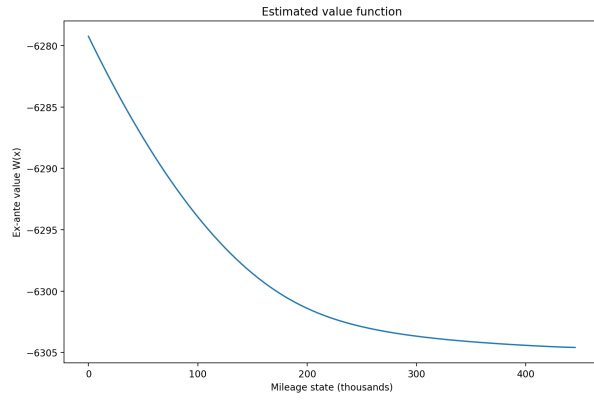
Notes: Dollar scale is computed as observed replacement cost divided by the estimated replacement-cost parameter. The final column is θ multiplied by this scale. The calculation is approximate and is used only to interpret the magnitude of the estimated cost slope.

The scaled estimates suggest that the implied maintenance cost rises faster for groups 1–3 than for group 4. This matches the qualitative interpretation in the original paper: newer buses are replaced at lower mileage even though their replacement cost is higher, which is consistent with steeper perceived operating cost increases.

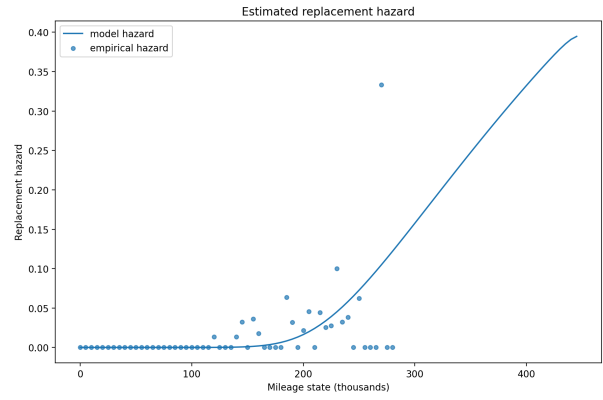
7 Model Fit: Value Functions, Hazards, and Equilibrium Distributions

7.1 Groups 1–3

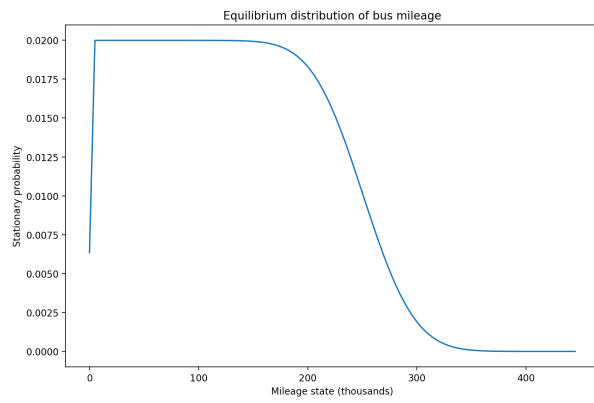
Figure 1 summarizes the fitted model for groups 1–3. The value function decreases with mileage because higher mileage raises expected future costs. The replacement hazard is close to zero at low mileage and rises strongly after roughly 200,000–250,000 miles. The stationary distribution places most probability mass below the high-mileage region where replacement becomes likely. The replacement demand function is downward sloping in RC .



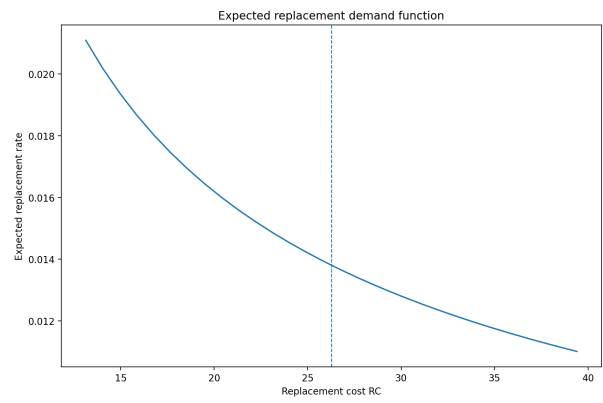
(a) Value function



(b) Replacement hazard



(c) Stationary mileage distribution

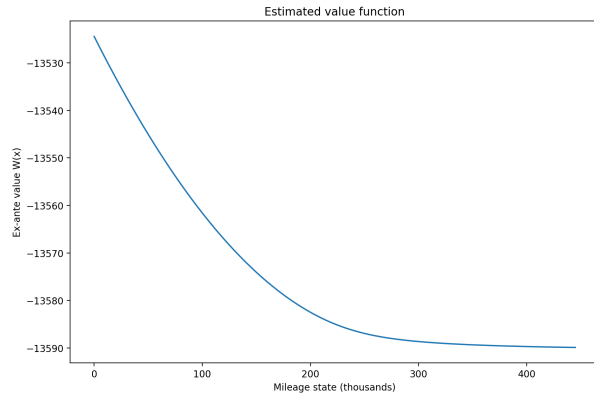


(d) Expected replacement demand

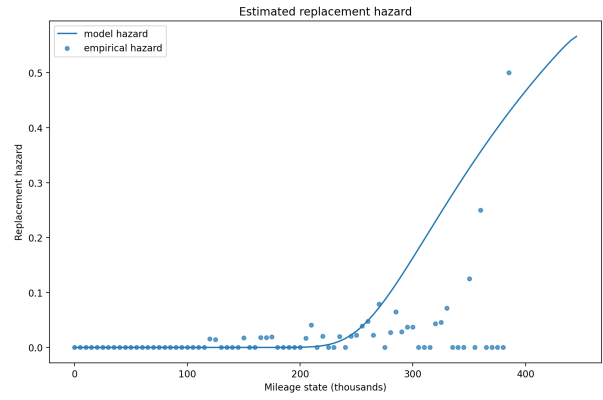
Figure 1: Model outputs for bus groups 1–3

7.2 Group 4

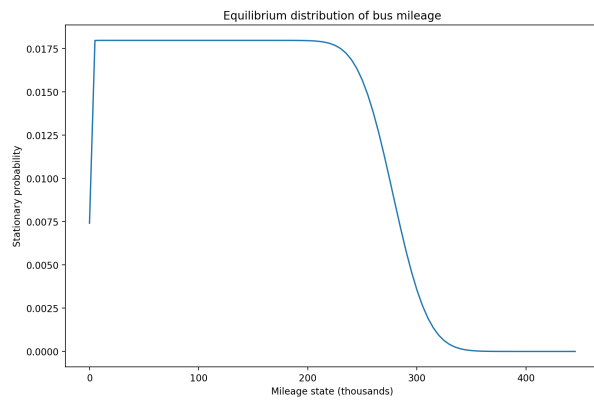
Figure 2 reports the corresponding objects for group 4. The estimated hazard for group 4 is shifted relative to groups 1–3. The stationary mean mileage is higher than for groups 1–3, and the expected monthly replacement rate is lower. This indicates that group 4 engines are kept longer on average in the fitted model.



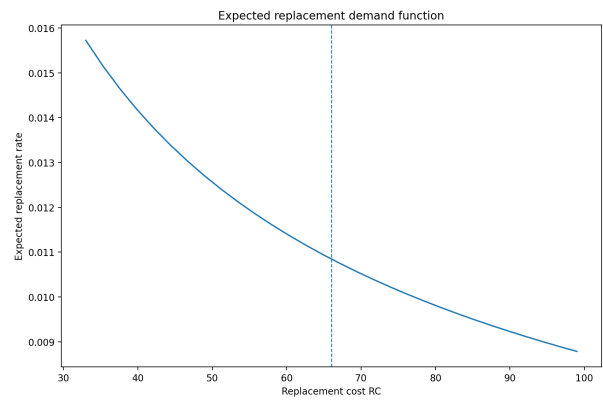
(a) Value function



(b) Replacement hazard



(c) Stationary mileage distribution

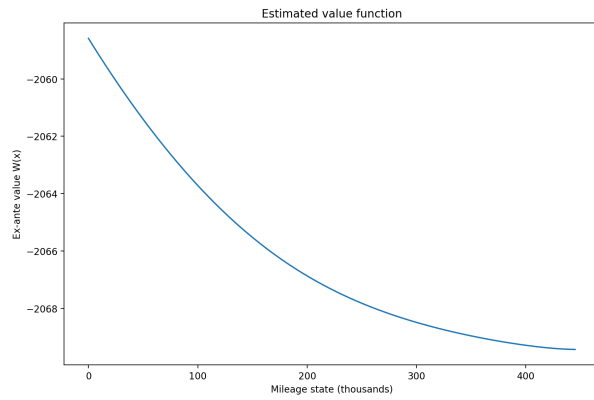


(d) Expected replacement demand

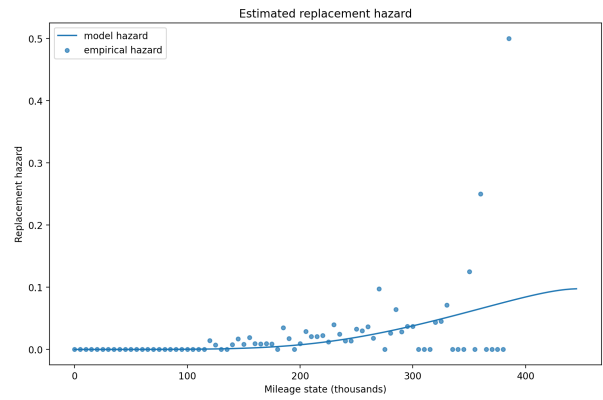
Figure 2: Model outputs for bus group 4

7.3 Groups 1–4 Pooled

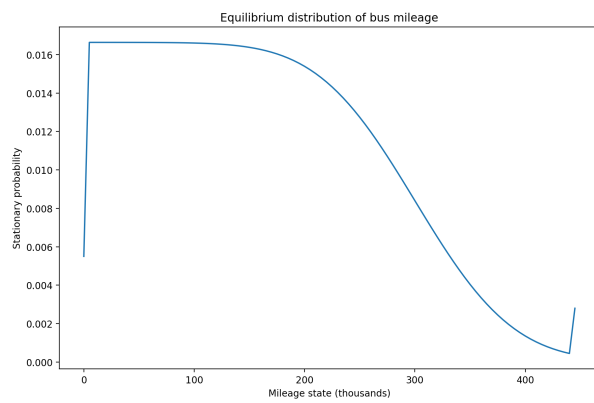
Figure 3 shows the pooled results. The pooled model imposes one common replacement-cost parameter, maintenance-cost slope, and transition process across all buses in groups 1–4. The resulting hazard is flatter than the separate-sample hazards. This flattening is important because it shifts the stationary distribution toward higher mileage and changes the counterfactual replacement response.



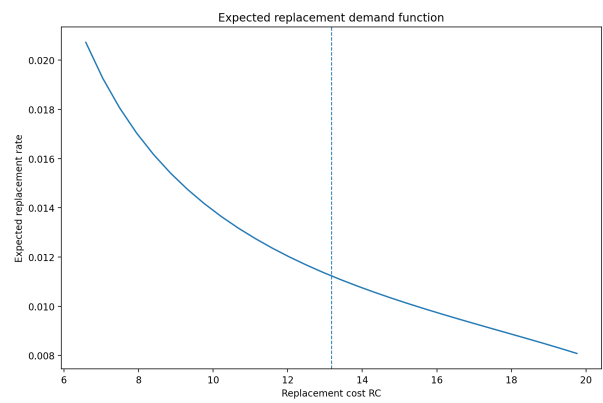
(a) Value function



(b) Replacement hazard



(c) Stationary mileage distribution



(d) Expected replacement demand

Figure 3: Model outputs for groups 1–4 pooled

8 Equilibrium Distribution

Table 3 reports summary moments of the stationary mileage distribution.

Table 3: Summary moments of the equilibrium distribution of bus mileage

Sample	Mean mileage	Mean state	SD state	Median state	Expected monthly replacement rate
Groups 1,2,3	128769.9	25.754	15.369	25	0.013806
Group 4	140671.2	28.134	16.456	28	0.010847
Groups 1–4 pooled	159471.0	31.894	19.976	30	0.011234

The stationary mean mileage is lowest for groups 1–3 and highest for the pooled sample. The pooled model has a heavier right tail because its replacement hazard is less steep at high mileage. This is a useful illustration of how pooling heterogeneous buses can change long-run predictions even when the model fits some average replacement behavior.

The expected monthly replacement rate is approximately 1.38% for groups 1–3, 1.08% for group 4, and 1.12% for the pooled sample. These rates are not simply sample replacement shares. They are stationary rates implied by the estimated dynamic policy and the controlled mileage transition process.

9 Counterfactual Replacement Demand

9.1 Counterfactual Design

The structural model can be used to simulate counterfactual replacement policies under alternative replacement costs. For each counterfactual value of RC , the replication recomputes the value function, the conditional replacement probabilities, the stationary mileage distribution, and the expected monthly replacement rate.

Let $D(RC)$ denote the expected replacement rate under replacement cost RC :

$$D(RC) = \sum_x \pi_{RC}(x) \mathbb{P}_{RC}(i = 1 \mid x).$$

Both terms depend on RC . A higher replacement cost directly lowers the current payoff from replacement and indirectly shifts the stationary distribution toward older engines. Thus, the demand curve is an equilibrium object, not a static choice probability.

9.2 Demand Grid Results

Table 4 reports selected points from the replacement-demand grid.

Table 4: Selected grid points from the expected replacement demand function

Sample	RC	$RC/\widehat{RC}$	Expected monthly replacement rate	Expected mileage
Groups 1,2,3	13.1344	0.500	0.021087	87037.6
	17.6635	0.672	0.017455	103350.3
	22.1927	0.845	0.015222	117417.9
	25.8159	0.983	0.013944	127558.5
	30.3451	1.155	0.012723	139198.0
	34.8742	1.328	0.011775	149935.9
	39.4033	1.500	0.011012	159955.4
Group 4	33.0039	0.500	0.015724	98044.3
	44.3846	0.672	0.013387	114548.0
	55.7652	0.845	0.011855	128957.6
	64.8697	0.983	0.010947	139418.2
	76.2504	1.155	0.010059	151483.1
	87.6310	1.328	0.009357	162657.5
	99.0117	1.500	0.008784	173114.2
Groups 1–4 pooled	6.5854	0.500	0.020724	96380.7
	8.8563	0.672	0.015412	121510.0
	11.1271	0.845	0.012753	142553.1
	12.9438	0.983	0.011378	157652.0
	15.2146	1.155	0.010113	175556.2
	17.4854	1.328	0.009087	194067.0
	19.7563	1.500	0.008080	216383.1

The demand curves are downward sloping. For groups 1–3, reducing RC to one-half of its estimated value raises the expected monthly replacement rate to 0.021087, while increasing RC to 1.5 times its estimated value lowers the replacement rate to 0.011012. At the same time, expected mileage rises from about 87,038 miles to about 159,955 miles. This is exactly the economic prediction of the model: higher replacement costs delay replacement and shift the fleet toward older engines.

For group 4, the same qualitative pattern appears. Increasing RC from 0.5 to 1.5 times its estimated value lowers the expected monthly replacement rate from 0.015724 to 0.008784. For the pooled model, the demand response is especially large in terms of expected mileage, which rises from about 96,381 miles to about 216,383 miles over the grid.

10 Specification Diagnostics

10.1 Lagged-Decision Dependence

Rust’s conditional independence assumption implies that, conditional on the current mileage state, lagged choices should not independently predict the current replacement decision. The simplified replication implements a compact diagnostic by estimating a one-parameter logit augmentation:

$$\log \left[\frac{\mathbb{P}(i_t = 1 \mid x_t, i_{t-1})}{1 - \mathbb{P}(i_t = 1 \mid x_t, i_{t-1})} \right] = \log \left[\frac{\hat{P}(i_t = 1 \mid x_t)}{1 - \hat{P}(i_t = 1 \mid x_t)} \right] + ai_{t-1}.$$

Under the null hypothesis of no lagged-decision dependence,

$$a = 0.$$

Table 5: Simplified LM-style diagnostic for lagged-decision dependence

Sample	LM statistic	df	p-value	$\hat{a}$	Test obs.
Groups 1,2,3	0.0000	1	0.999959	-4.8485	3864
Group 4	0.0000	1	0.999954	-5.2786	4292
Groups 1–4 pooled	0.0001	1	0.991469	-10.0000	8156

Notes: This is a simplified fitted-probability-offset diagnostic. It is motivated by Rust’s conditional independence test but should not be interpreted as an exact replication of Rust’s Table XI.

The diagnostic results should be interpreted cautiously. The estimated values of $\hat{a}$ are

negative, and the pooled estimate reaches the numerical lower bound. This suggests that lagged replacement is sparse and that the simplified test may be numerically fragile. Therefore, the table is best viewed as a diagnostic check rather than a formal specification test.

10.2 Heterogeneity Diagnostic

The replication also computes a likelihood-ratio comparison between a restricted pooled model and an unrestricted model that estimates groups 1–3 and group 4 separately. The likelihood-ratio statistic is

$$LR = 2[\ell_{\text{unrestricted}} - \ell_{\text{restricted}}].$$

Table 6: Likelihood-ratio heterogeneity diagnostic

Restricted model	Unrestricted model	LR statistic	df	p-value	LL restricted	LL unrestricted
Groups 1–4 pooled	Groups 1,2,3 + Group 4	-69.2665	4	1.000000	-6112.062	-6146.695

The negative likelihood-ratio statistic is not theoretically valid. Since the pooled model is nested within the separate-sample model, the unrestricted likelihood should be weakly larger than the restricted likelihood. The negative statistic therefore indicates a computational issue, most likely local convergence or imperfect comparability of likelihood evaluations across runs. For this reason, this table should be treated as a warning flag rather than final evidence against heterogeneity.

11 Discussion

The simplified replication captures the central qualitative implications of Rust’s dynamic replacement model. First, the value function decreases with mileage. This reflects the fact that high-mileage engines have higher current and future expected operating costs. Second, replacement hazards increase with mileage. Replacement is unlikely for low-mileage engines but becomes more likely as accumulated mileage rises. Third, the counterfactual replacement demand function is downward sloping in the replacement-cost parameter. Higher replacement costs make replacement less attractive and shift the stationary fleet distribution toward older engines.

The results also show why bus-group heterogeneity matters. Groups 1–3 and group 4 have different estimated replacement-cost parameters, maintenance-cost slopes, and transition probabilities. The pooled model compresses these differences into one set of com-

mon parameters. As a result, the pooled hazard is flatter and the stationary distribution has a heavier right tail. This changes the implied long-run mileage distribution and the replacement-cost counterfactuals.

At the same time, this replication remains simplified relative to the original paper. First, only the linear maintenance-cost specification is estimated. Rust’s paper also studies alternative functional forms, including square-root, quadratic, cubic, hyperbolic, mixed, and nonparametric cost specifications. Second, the current standard errors are based on the optimizer’s approximate inverse Hessian and should be improved using a more reliable numerical Hessian or outer-product-of-gradients matrix. Third, the diagnostic tests are simplified and should not be interpreted as exact replications of Rust’s formal specification tests. Fourth, the negative likelihood-ratio statistic indicates that the separate-sample optimization should be debugged before reporting a formal heterogeneity test.

Despite these limitations, the replication demonstrates the main computational and economic logic of the nested fixed point approach. The model estimates primitive parameters, solves the dynamic program, constructs policy functions, and then uses those policy functions to make equilibrium and counterfactual predictions.

12 Conclusion

This paper replicates the core structure of Rust’s bus engine replacement model using a simplified Python implementation of the nested fixed point algorithm. The model estimates a replacement-cost parameter, a linear maintenance-cost slope, and discrete mileage transition probabilities. It solves the Bellman equation, computes dynamic replacement probabilities, constructs stationary mileage distributions, and simulates counterfactual replacement demand under alternative replacement costs.

The main findings are consistent with dynamic optimal replacement theory. Higher mileage lowers the value of keeping an engine and raises the probability of replacement. Higher replacement costs reduce the expected monthly replacement rate and increase the expected mileage of the fleet. The separate-sample results suggest that replacement behavior differs across bus groups, while the pooled model smooths over these differences and changes the long-run predictions.

The replication provides a transparent foundation for further extensions. Future work should improve the optimizer diagnostics, compute more reliable standard errors, implement Rust’s exact specification tests, compare alternative cost functions, and verify the heterogeneity likelihood-ratio test using multiple starting values and a common likelihood-evaluation routine.

A Implementation Details

The Python replication is organized into several blocks. The model specification sets the number of states to 90, the mileage bin size to 5,000 miles, and the discount factor to $\beta = 0.9999$. The data preparation routine builds the panel for each sample, re-indexes bus identifiers, clips mileage increments into the three categories $\{0, 1, 2\}$, and top-codes the mileage state at 89.

The function that solves the dynamic programming problem constructs the keep transition matrix, computes the linear maintenance-cost vector, and iterates on the log-sum Bellman operator. The choice-probability function then computes dynamic logit keep and replacement probabilities. The likelihood function sums the log probabilities of the observed choices and the log probabilities of the observed mileage increments.

After estimation, the code constructs the policy-induced transition matrix, computes its stationary distribution, and obtains the expected monthly replacement rate. The counterfactual demand routine varies RC over a grid from 0.5 to 1.5 times the estimated value, recomputes the value function and policy at each grid point, and records the resulting stationary replacement rate and expected mileage.

The figure routines produce four outputs for each sample:

1. the estimated value function;
2. the model-implied and empirical replacement hazard;
3. the equilibrium mileage distribution;
4. the expected replacement demand function.

The figures in the main text are arranged as 2-by-2 panels to keep each image inside the page while preserving readability.

References

Rust, John. 1987. Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher. *Econometrica*, 55(5): 999–1033.